

Digestion, Liver

1. What are the end products of carbohydrate, fat and protein digestion?

Digestive end products of

1. Carbohydrates are monosaccharides ; mainly glucose and others such as fructose, galactose
2. Lipids are free fatty acids, monoacyl glycerol, lysophospholipid, cholesterol
3. Proteins are amino acids.

2. Enumerate the enzymes of pancreatic origin.

Enzymes of pancreatic origin

- (1) Enzyme for carbohydrate digestion-pancreatic alpha amylase
- (2) Enzymes for protein digestion-;trypsin elastase carboxypeptidase
- (3) Enzymes for lipids digestion- pancreatic lipase and colipase pancreatic phospholipase A2 pancreatic cholesterol esterase.

3. What are Xenobiotics ?

Xenobiotics are various foreign chemicals which are exposed by humans.

Xenobiotics are toxic substances such as drugs, insecticides, pollutants, food additives that are harmful to the body.

4. Write a brief account of hydroxylation reaction?

Hydroxylation

This is the major reaction, catalyzed by cytochrome P₄₅₀ monooxygenase. It requires NADPH and molecular oxygen.



RH can represent drugs, carcinogens, pesticides, petroleum products, and endogenous compounds such as steroids, eicosanoids and retinoids.

5. Enumerate the conjugation reactions involved in detoxification functions.

- a. Conjugation with glucuronic acid
- b. Conjugation with glutathione
- c. Conjugation with glycine
- d. Conjugation with sulfate(Sulfation)
- e. Conjugation with acetate(Acetylation)
- f. Methylation

6. What are the function of liver?

1. Circulatory function
2. Haematologic function
3. Storage of vitamins and minerals
4. Immune regulation
5. Biotransformation function
6. Detoxification function of liver.

7. Explain Biotransformation functions?

Biotransformation function occurring in the liver are vital functions.

- (a) Synthesis of bile acids from cholesterol
- (b) Synthesis of bile pigment
- (c) Hormonal metabolism
- (d) Synthesis of calcitriol (1,25 DHCC)
- (e) Synthesis of ammonia
- (f) Synthesis of purine and pyrimidine basis

8. What are bile salts?

Bile salts are Na^+ and K^+ salts of bile acids conjugated to glycine and taurine. The primary bile acids are cholic acid and chenodeoxycholic acid synthesized from cholesterol in the liver. Secondary bile acids are deoxycholic acid and lithocholic acid formed by bacterial action in the intestine.

9. How are primary bile acids formed in the liver?

The primary bile acids are cholic and chenodeoxycholic acid. They are synthesized in the liver from cholesterol. Cholesterol is converted to 7_α – hydroxyl cholesterol catalyzed by 7_α – hydroxylase (requires O_2 , NADPH and cytochrome P_{450}). It is the rate limiting step and requires vitamin C. After that the process is divided into 2 pathways;

- (1) For synthesis of cholic acid
- (2) For synthesis of chenodeoxycholic acid pathway.

Conjugation of CoA derivatives with glycine or taurine give glycocholic acid or glychenodeoxycholic acid and taurocholic acid or taurochenodeoxycholic acid. The major conjugated bile acid is glycocholic acid.

10. State the functions of bile salts.

“Functions of bile salts”

In the liver –

- (a) Micelle formation for cholesterol excretion
- (b) Inhibition of hepatic cholesterol synthesis
- (c) Stimulation of cheleresis

In the gut lumen –

- (a) Emulsification of lipids for digestion
- (b) Micelle formation of lipids for absorption
- (c) Activation of enzymes
- (d) Absorption of fat soluble vitamins

In the mucosal cell—

- (a) Stimulation of re-esterification
- (b) Stimulation of lipoprotein formation(chylomicron formation)
- (c) Inhibition of cholesterol synthesis
- (d) Stimulation of cholecystokinin (CCK) release